AAE57500 Introduction to Satellite Navigation and Positioning
3 credits

1 Course Information/Description

Introduction to radio-navigation techniques, using the Global Navigation Satellite System (GNSS); GNSS signal structures; satellite search and acquisition; satellite tracking; coordinate systems and time; observations; atmospheric effects; and position-velocity-time (PVT) solutions.

2 Learning Resources and Reference Text

Brightspace Learning Management System:

Access the course videos, materials, and resources via Purdue's Brightspace learning management system.

Required Textbook:

Position, Navigation, and Timing Technologies in the 21st Century: Integrated Satellite Navigation, Sensor Systems, and Civil Applications, T. Jade Morton, Frank van Diggelen, James J. Spilker Jr., Bradford W. Parkinson, Sherman Lo, Grace Gao, eds. First published:15 December 2020, IEEE, DOI:10.1002/9781119458555

Available as e-book through Purdue Libraries.

Optional Textbook:

Global Positioning System: Signals, Measurements and Performance, Misra, 2nd edition, ISBN 9780970954411

Additional Readings:

There will be additional readings for some of the sub-sections. All will be distributed through links to the specific module on the Brightspace page at the needed time.

Software/web Resources:

Students are expected to have access to and be familiar with the standard software used in engineering, including word processing, spreadsheets, presentation (e.g. Powerpoint) and computation (e.g MATLAB). Some example code and useful functions will be provided in MATLAB.

3 Assignments

3.1 Project

Each student will be required to complete a comprehensive literature review on a topic of their choice, selected in consultation with the instructor. If possible, students are advised to find a project

relevant to their job. Students are also expected to make some original contribution as part of their project. Examples of an original contribution include: independent reproduction of a published result; comparison of different approaches to solving a problem; or applying a published method to a simple example problem. The project has four deliverables with due dates spaced at approximately equal intervals through the semester: (1) Abstract, (2) Literature review, (3) Draft report, (4) Final report. Timely feedback will be given for (1) (2) and (3), if they are submitted by the due date. Only the final report (4) will be used to compute the final grade.

3.2 Homework

5 to 6 homework assignments will be given. Homework assignments will involve a substantial amount of computer programming and data processing. as a single PDF, giving a narrative describing the problem and the student's approach to solving it. All homework will be submitted through Brightspace.

Each student team will complete the design of a satellite mission to meet specific requirements provided by the instructor. Feedback from the mid-term (Blue Team) and final (Red Team) reviews is intended for the teams to use in identifying any problems with their design early in the process and allow them the opportunity to correct them. A written proposal, following an industry standard format will be submitted and a final presentation will be given by each team.

4 Course Learning Objectives

1. Understand how basic physical principles (eg., orbits, reference frames, electromagnetic propagation, atmospheric effects and statistical estimation) are integrated in the Global Navigation Satellite System (GNSS)

2. State mathematical models for the basic GNSS observations: pseudorange, carrier phase and Doppler and understand the characteristics of errors and biases in these observations.

3. Create an algorithm for producing pseudorandom noise (PRN) codes and demonstrate its use in detecting individual transmitters in the signal received by a GNSS receiver.

4. Calculate corrections for atmospheric effects on the basic GNSS observations using models and multiple frequencies.

5. Calculate the GNSS satellite orbits from the broadcast ephemeris

6. Solve for the position of a GNSS receiver from a set of 5 or more pseudorange observations and predict the geometric dilution of precision (GDOP) for this solution.

7. Demonstrate the use of a double-difference method in solving for the baseline between two receivers, using carrier phase observations.

8. Perform a literature review and write a scientific article.

5 Evaluation

Final grades should measure each student's level of understanding of the course material. Letter grades will be determined based upon a pre-defined scale (not relative to other students).

Homework: 50%

Final Project: 50%

The minimum requirement for a "B" on the final project is a thorough review of current research literature on a topic selected by the student. Some individual contribution is required for the "A" grade. For example; Applying a published algorithm to a simple problem, Reproducing published results, comparing multiple approaches to a problem

The final numerical score will be directly converted to a letter grade according to the intervals in the following table:

A+: >=97
A: [93, 97)
A-: [90, 93)
B+: [87, 90)
B: [83, 87)
B-: [80, 83)
C+: [77, 80)
C: [73, 77)
C-: [70, 73)
D+: [67, 70)
D: [63, 67)
D-: [60, 63)
F: <60